Week 07 teacher guide

Area and surface area

Keep this guide beside PowerPoint slides 377-438. Follow the numbered slide sequence, then use the matching answer section after the student has tried.

The seven days

Day 043. Saturday | pages 166-171 | slides 377-394
Area from a base and height

Day 044. Sunday | pages 172-184 | slides 395-414
Nets and surface area

Day 045. Monday | pages 185-186
Area and surface area

Day 046. Tuesday | pages 187-188
Area and surface area

Day 047. Wednesday | pages 189-190
Area and surface area

Day 048. Thursday | pages 191-192
Area and surface area

Day 049. Friday | pages 193-194
Weekly check

This week's idea

Area counts square units that cover a surface. Cutting and rearranging reveal triangle and parallelogram formulas. A net helps count every face of a solid.

Use the built-in Booster review where it appears in the day's work. It replaces ordinary practice. No separate Booster packet is needed.

PDF bookmarks jump to each lesson and answer section. The student workbook's Sunday reference pages are for checking an example after an attempt. Keep this teacher guide and the answer-review slides out of view during independent work.

Saturday. Area from a base and height

Day 043, pages 166-171 | PowerPoint slides 377-394

Goal: Find area by matching a base with its perpendicular height.

Set out: grid paper, scissors, ruler, paper triangle.

Check first: Rectangle area and multiplication with fractions.

Pacing: Retrieval 7; launch 8; explore 20; discuss 15; synthesis 7; exit 8; preview 5; slack 5. Total 75 min.

Explore: what to watch for

Marks a right angle at the foot of the height. Why it matters: Identifies the distance the formula needs. Tutor move: Ask whether the slanted side makes that angle.

Rearranges without gaps or overlaps. Why it matters: Shows why area is preserved. Tutor move: Compare the original and rearranged boundaries.

Chooses either leg of a right triangle as base. Why it matters: Different valid choices give the same area. Tutor move: Accept both and compare base-height pairs.

Likely misconceptions

Uses a slanted side as height. Why: Treats any given length as interchangeable. Tutor move: Trace the perpendicular distance with a ruler.

Omits the half for a triangle. Why: Uses the containing parallelogram. Tutor move: Pair congruent copies and split the area equally.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 377. Week 7, Saturday

Topic: Area from a base and height
Student: Watch and discuss

Say: Find area by matching a base with its perpendicular height.

Ask: What do you already know about this idea?

Expect: Rectangle area and multiplication with fractions.

If stuck: Ask whether the slanted side makes that angle.

Slide 378. Today

Topic: Area from a base and height
Student: Watch and discuss

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: A parallelogram rearranges into a rectangle. Two copies of a triangle make a parallelogram.

If stuck: Read the learning goal together.

Slide 379. Recall | 4 min

Topic: Area from a base and height
Student: Day 043 · Workbook p. 166 · R1, R2, R3, R4

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: 3 $\frac{1}{3}$; 2: $7.35; 3: 9; 4: 6

If stuck: Give time to draw before giving a hint.

Slide 380. Recall: answers | 3 min

Topic: Area from a base and height
Student: Day 043 · Workbook p. 166 · R1, R2, R3, R4

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: 3 $\frac{1}{3}$; $7.35; 9; 6

If stuck: Rebuild one missed item with its model.

Slide 381. Rectangle area review | 8 min

Topic: Area from a base and height
Student: Day 043 · Workbook p. 167 · P1

Say: Read P1. Give time to try before comparing methods.

Ask: P1: Are we measuring the surface inside or the distance around the shape?

Expect: P1: Area 15 square cm; perimeter 17 cm

If stuck: P1: Cover the shape with square units for area. Trace the edge separately for perimeter.

Slide 382. Parallelogram | 3 min

Topic: Area from a base and height
Student: Day 043 · Workbook p. 167 · P2

Say: Read P2. Give time to try before comparing methods.

Ask: P2: Which length is perpendicular to the chosen base?

Expect: P2: 28 square cm

If stuck: P2: Mark the right angle. Move an end triangle to make a rectangle with the same base and height.

Slide 383. The model | 3 min

Topic: Area from a base and height
Student: Watch and discuss

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: Cut and slide the left end to make a rectangle.

If stuck: Ask whether the slanted side makes that angle.

Slide 384. Right triangle | 3 min

Topic: Area from a base and height
Student: Day 043 · Workbook p. 167 · P3

Say: Read P3. Give time to try before comparing methods.

Ask: P3: Which base-height pair makes a right angle? How is the triangle related to a parallelogram?

Expect: P3: 24 square cm both ways

If stuck: P3: Pair two copies of the triangle. Find their combined area, then split it equally.

Slide 385. Height is not a side | 3 min

Topic: Area from a base and height
Student: Day 043 · Workbook p. 168 · P4

Say: Read P4. Give time to try before comparing methods.

Ask: P4: Which base-height pair makes a right angle? How is the triangle related to a parallelogram?

Expect: P4: 60 square cm

If stuck: P4: Pair two copies of the triangle. Find their combined area, then split it equally.

Slide 386. Obtuse triangle | 2 min

Topic: Area from a base and height
Student: Day 043 · Workbook p. 168 · P5

Say: Read P5. Give time to try before comparing methods.

Ask: P5: Which base-height pair makes a right angle? How is the triangle related to a parallelogram?

Expect: P5: 6 square cm

If stuck: P5: Pair two copies of the triangle. Find their combined area, then split it equally.

Slide 387. Rational dimensions | 2 min

Topic: Area from a base and height
Student: Day 043 · Workbook p. 168 · P6

Say: Read P6. Give time to try before comparing methods.

Ask: P6: Which base-height pair makes a right angle? How is the triangle related to a parallelogram?

Expect: P6: 6 square cm; no, the height must be perpendicular

If stuck: P6: Pair two copies of the triangle. Find their combined area, then split it equally.

Slide 388. Watch the thinking | 2 min

Topic: Area from a base and height
Student: Watch and discuss · worked example

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 18 square cm

If stuck: Return to the matching unit in the model.

Slide 389. Your turn | 2 min

Topic: Area from a base and height
Student: Day 043 · Workbook p. 169 · F

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 2.5; 10 square cm

If stuck: Compare with the previous example without copying its numbers.

Slide 390. A claim to check | 15 min

Topic: Area from a base and height
Student: Day 043 · Workbook p. 169 · D

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: The triangle is half the parallelogram: 18 square cm.

If stuck: Place two matching paper triangles together.

Slide 391. In your own words | 7 min

Topic: Area from a base and height
Student: Day 043 · Workbook p. 166 · S

Say: Use your own words and one example.

Ask: Partner compare: Why does moving a cut piece preserve area?

Expect: A parallelogram rearranges into a rectangle. Two copies of a triangle make a parallelogram.

If stuck: Point to the model and name what each part represents.

Slide 392. Exit ticket | 8 min

Topic: Area from a base and height
Student: Day 043 · Workbook p. 170 · E

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: 48 square cm; height 8 cm

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 393. The week ahead | 5 min

Topic: Area from a base and height
Student: Days 045-049 · Workbook pp. 185-194

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 394. Tutor reference

Topic: Area from a base and height

Student: Tutor only · hidden in slide show

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: 48 square cm; height 8 cm

Got it: Gives 48 square cm and identifies 8 cm as perpendicular height.

Almost there: Uses $\frac{1}{2} \times 12 \times 8$ but makes one arithmetic or unit slip.

Needs re-teaching: Uses 10 as height, omits the half, or cannot identify a perpendicular height.

Sunday. Nets and surface area

Day 044, pages 172-184 | PowerPoint slides 395-414

Goal: Use decompositions and nets to count every part of a surface.

Set out: paper nets, scissors, tape, ruler.

Check first: Rectangle and triangle area; multiplication and addition.

Pacing: Retrieval 5; launch 5; explore 15; discuss 12; synthesis 5; mixed review 10; exit 8; preview 5. Total 65 min.

Explore: what to watch for

Labels both hidden and visible faces. Why it matters: Prevents missing areas. Tutor move: Match each label on the solid and the net.

Uses different decompositions for the L. Why it matters: Area is independent of a valid split. Tutor move: Accept both and compare the totals.

Uses square units for faces and cubic units for volume. Why it matters: Distinguishes surface from capacity. Tutor move: Ask what one unit would look like.

Likely misconceptions

Uses length × width × height for surface area. Why: Confuses space inside with covering outside. Tutor move: Imagine covering every face with paper.

Counts only one triangular end. Why: A sketch hides the other end. Tutor move: Fold the net and point to both ends.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 395. Week 7, Sunday

Topic: Nets and surface area
Student: Watch and discuss

Say: Use decompositions and nets to count every part of a surface.

Ask: What do you already know about this idea?

Expect: Rectangle and triangle area; multiplication and addition.

If stuck: Match each label on the solid and the net.

Slide 396. Today

Topic: Nets and surface area
Student: Watch and discuss

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: A net unfolds every face. Surface area adds face areas in square units.

If stuck: Read the learning goal together.

Slide 397. Recall | 3 min

Topic: Nets and surface area
Student: Day 044 · Workbook p. 172 · R1, R2, R3, R4

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: 2 $\frac{1}{2}$; stories vary; 2: 7.714; 3: x = 14; 4: 1 meters

If stuck: Give time to draw before giving a hint.

Slide 398. Recall: answers | 2 min

Topic: Nets and surface area
Student: Day 044 · Workbook p. 172 · R1, R2, R3, R4

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: 2 $\frac{1}{2}$; stories vary; 7.714; x = 14; 1 meters

If stuck: Rebuild one missed item with its model.

Slide 399. L-shaped polygon split two ways | 5 min

Topic: Nets and surface area
Student: Day 044 · Workbook p. 173 · P1

Say: Read P1. Give time to try before comparing methods.

Ask: P1: Which simple regions cover the shape without gaps or overlaps?

Expect: P1: 42 square cm

If stuck: P1: Draw one straight cut, or complete a larger rectangle and identify the part to remove.

Slide 400. Rectangular prism net | 3 min

Topic: Nets and surface area
Student: Day 044 · Workbook p. 173 · P2

Say: Read P2. Give time to try before comparing methods.

Ask: P2: Which faces need covering, including the hidden faces?

Expect: P2: Surface area 52 square cm; volume 24 cubic cm

If stuck: P2: Label every face on a net. Find each face area before adding the complete list.

Slide 401. The model | 2 min

Topic: Nets and surface area
Student: Watch and discuss

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: Match each rectangle to one face.

If stuck: Match each label on the solid and the net.

Slide 402. Triangular prism net | 2 min

Topic: Nets and surface area
Student: Day 044 · Workbook p. 173 · P3

Say: Read P3. Give time to try before comparing methods.

Ask: P3: Which faces need covering, including the hidden faces?

Expect: P3: Two triangles of area 24 each; rectangles of area 30, 40, 50; total 168 square cm

If stuck: P3: Label every face on a net. Find each face area before adding the complete list.

Slide 403. Square pyramid net | 2 min

Topic: Nets and surface area
Student: Day 044 · Workbook p. 174 · P4

Say: Read P4. Give time to try before comparing methods.

Ask: P4: Which faces need covering, including the hidden faces?

Expect: P4: 96 square cm

If stuck: P4: Label every face on a net. Find each face area before adding the complete list.

Slide 404. Find a missing face in a surface-area calculation | 2 min

Topic: Nets and surface area
Student: Day 044 · Workbook p. 174 · P5

Say: Read P5. Give time to try before comparing methods.

Ask: P5: Which faces need covering, including the hidden faces?

Expect: P5: 76 square cm

If stuck: P5: Label every face on a net. Find each face area before adding the complete list.

Slide 405. Watch the thinking | 2 min

Topic: Nets and surface area
Student: Watch and discuss · worked example

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 32 square cm

If stuck: Return to the matching unit in the model.

Slide 406. Your turn | 2 min

Topic: Nets and surface area
Student: Day 044 · Workbook p. 174 · F

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 3; 40 square cm

If stuck: Compare with the previous example without copying its numbers.

Slide 407. Two ways to see it | 6 min

Topic: Nets and surface area

Student: Watch and discuss · compare the same example

Say: Partner compare: show the same problem in two ways.

Ask: How does each part of the model match a step in the calculation? Partner compare: Where is every part of the L in each split? Agree or disagree: Does a hidden face still need wrapping paper?

Expect: Accept drawings and equations that preserve the same quantities and units. Uses different decompositions for the L.

If stuck: Accept both and compare the totals.

Slide 408. A claim to check | 6 min

Topic: Nets and surface area

Student: Day 044 · Workbook p. 175 · D

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: A closed cube has six faces: $6 \times 9 = 54$ square cm.

If stuck: Label each face of the folded cube and net with matching letters.

Slide 409. Words and meaning | 3 min

Topic: Nets and surface area

Student: Watch and discuss

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: An L can split into two rectangles. Six rectangles can form a box. A closed box includes its bottom. Volume uses cubic units.

If stuck: Read one definition and draw its example.

Slide 410. Mixed review | 10 min

Topic: Nets and surface area

Student: Day 044 · Workbook p. 176 · M1, M2, M3

Say: Choose a method for each problem.

Ask: What tells you which operation to use?

Expect: 12 pieces; 13; (1, 2), (3, 6), (5, 10)

If stuck: Name the quantities and units before calculating.

Slide 411. In your own words | 2 min

Topic: Nets and surface area
Student: Day 044 · Workbook p. 172 · S

Say: Use your own words and one example.

Ask: Partner compare: Where is every part of the L in each split?

Expect: A net unfolds every face. Surface area adds face areas in square units.

If stuck: Point to the model and name what each part represents.

Slide 412. Exit ticket | 8 min

Topic: Nets and surface area
Student: Day 044 · Workbook p. 177 · E

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: 108 square cm

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 413. The week ahead | 5 min

Topic: Nets and surface area
Student: Days 045-049 · Workbook pp. 185-194

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 414. Tutor reference

Topic: Nets and surface area
Student: Tutor only · hidden in slide show

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: 108 square cm

Got it: Gives 108 square cm and includes two triangles and three rectangles.

Almost there: Lists all five faces correctly but makes one area or addition slip.

Needs re-teaching: Leaves out a face, uses volume, or doubles a triangular face area.

PowerPoint answer review

After Friday, outside the weekend lesson time. Say: Check one completed item at a time. Ask: Where does your method first differ from this one? Expect: The answers and worked reasoning printed in the complete keys below. If stuck: Rebuild the quantity with its model, then have the student repair the first incorrect step.

Keep these slides out of view during independent work. The index below follows the answer-review slides in order; every answer is also printed in the complete keys.

Slide 415. Week 7 answers

Next lesson: slide 439. Use this review section after Friday.

Slide 416. Area from a base and height (1/4)

Items: S13:R1; S13:R2; S13:R3; S13:R4. Follow along: Day 043 · Workbook p. 166 · Items R1, R2, R3, R4

Slide 417. Area from a base and height (2/4)

Items: S13:S; S13:P1; S13:P2; S13:P3. Follow along: Day 043 · Workbook pp. 166, 167 · Items S, P1, P2, P3

Slide 418. Area from a base and height (3/4)

Items: S13:P4; S13:P5; S13:P6; S13:F; S13:D. Follow along: Day 043 · Workbook pp. 168, 169 · Items P4, P5, P6, F, D

Slide 419. Area from a base and height (4/4)

Items: S13:E. Follow along: Day 043 · Workbook p. 170 · Items E

Slide 420. Nets and surface area (1/4)

Items: S14:R1; S14:R2; S14:R3; S14:R4. Follow along: Day 044 · Workbook p. 172 · Items R1, R2, R3, R4

Slide 421. Nets and surface area (2/4)

Items: S14:S; S14:P1; S14:P2. Follow along: Day 044 · Workbook pp. 172, 173 · Items S, P1, P2

Slide 422. Nets and surface area (3/4)

Items: S14:P3; S14:P4; S14:P5; S14:F. Follow along: Day 044 · Workbook pp. 173, 174 · Items P3, P4, P5, F

Slide 423. Nets and surface area (4/4)

Items: S14:D; S14:M1; S14:M2; S14:M3; S14:E. Follow along: Day 044 · Workbook pp. 175, 176, 177 · Items D, M1, M2, M3, E

Slide 424. Study guide: Try it

Items: W07_Study_Try:1; W07_Study_Try:2; W07_Study_Try:3; W07_Study_Try:4; W07_Study_Try:5. Follow along: Day 044 · Workbook p. 181 · Items 1, 2, 3, 4, 5

Slide 425. Daily practice (1/3)

Items: Day045_Practice:1; Day045_Practice:2; Day045_Practice:3; Day045_Practice:4; Day045_Practice:5. Follow along: Day 045 · Workbook p. 185 · Items 1, 2, 3, 4, 5

Slide 426. Daily practice (2/3)

Items: Day045_Practice:6; Day045_Practice:7; Day045_Practice:8; Day045_Practice:9; Day045_Practice:10. Follow along: Day 045 · Workbook pp. 185, 186 · Items 6, 7, 8, 9, 10

Slide 427. Daily practice (3/3)

Items: Day045_Practice:11; Day045_Practice:12; Day045_Practice:13. Follow along: Day 045 · Workbook p. 186 · Items 11, 12, 13

Slide 428. Daily practice (1/3)

Items: Day046_Practice:1; Day046_Practice:2; Day046_Practice:3; Day046_Practice:4; Day046_Practice:5. Follow along: Day 046 · Workbook p. 187 · Items 1, 2, 3, 4, 5

Slide 429. Daily practice (2/3)

Items: Day046_Practice:6; Day046_Practice:7; Day046_Practice:8; Day046_Practice:9; Day046_Practice:10. Follow along: Day 046 · Workbook pp. 187, 188 · Items 6, 7, 8, 9, 10

Slide 430. Daily practice (3/3)

Items: Day046_Practice:11; Day046_Practice:12; Day046_Practice:13. Follow along: Day 046 · Workbook p. 188 · Items 11, 12, 13

Slide 431. Daily practice (1/3)

Items: Day047_Practice:1; Day047_Practice:2; Day047_Practice:3; Day047_Practice:4; Day047_Practice:5. Follow along: Day 047 · Workbook p. 189 · Items 1, 2, 3, 4, 5

Slide 432. Daily practice (2/3)

Items: Day047_Practice:6; Day047_Practice:7; Day047_Practice:8; Day047_Practice:9; Day047_Practice:10. Follow along: Day 047 · Workbook pp. 189, 190 · Items 6, 7, 8, 9, 10

Slide 433. Daily practice (3/3)

Items: Day047_Practice:11; Day047_Practice:12; Day047_Practice:13. Follow along: Day 047 · Workbook p. 190 · Items 11, 12, 13

Slide 434. Daily practice (1/3)

Items: Day048_Practice:1; Day048_Practice:2; Day048_Practice:3; Day048_Practice:4; Day048_Practice:5. Follow along: Day 048 · Workbook p. 191 · Items 1, 2, 3, 4, 5

Slide 435. Daily practice (2/3)

Items: Day048_Practice:6; Day048_Practice:7; Day048_Practice:8; Day048_Practice:9; Day048_Practice:10. Follow along: Day 048 · Workbook pp. 191, 192 · Items 6, 7, 8, 9, 10

Slide 436. Daily practice (3/3)

Items: Day048_Practice:11; Day048_Practice:12; Day048_Practice:13; Day048_Practice:14. Follow along: Day 048 · Workbook p. 192 · Items 11, 12, 13, 14

Slide 437. Weekly check (1/2)

Items: Day049_Practice:1; Day049_Practice:2; Day049_Practice:3; Day049_Practice:4; Day049_Practice:5. Follow along: Day 049 · Workbook p. 193 · Items 1, 2, 3, 4, 5

Slide 438. Weekly check (2/2)

Items: Day049_Practice:6; Day049_Practice:7; Day049_Practice:8; Day049_Practice:9; Day049_Practice:10. Follow along: Day 049 · Workbook p. 194 · Items 6, 7, 8, 9, 10

Weekday teaching and review

Use one day at a time. Aim for 15-20 minutes, or 15 minutes during a holiday week. Accuracy and a clear explanation matter more than finishing quickly.

Day 045. Monday | pages 185-186

Ask for a model from Saturday before checking an answer. Let the student choose a method. After checking, have the student correct the first step that changed the quantity.

Built-in Booster review: Area, surface area and volume: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 046. Tuesday | pages 187-188

Connect the practice to Sunday's model. Ask what each number counts and how a multiplication check, diagram, or estimate supports the answer.

Built-in Booster review: Area, surface area and volume: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 047. Wednesday | pages 189-190

Let the student choose the method for each mixed problem. For the puzzle, ask which conditions are already met. Accept any solution that meets every condition.

Built-in Booster review: Weekly puzzle: items 13. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 048. Thursday | pages 191-192

Ask the student to explain the planted error before repairing it. Optional Stretch comes after the required work and may be skipped without delaying the next day.

Built-in Booster review: Expressions and equations: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 049. Friday | pages 193-194

Give the weekly check independently, without coaching or a calculator. Score the required reasoning as well as the answer. Afterward, choose one error to discuss calmly.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

After Friday

On a 10-item check: 9-10 correct is strong; 7-8 is on track; 5-6 needs another review of the missed topic; 0-4 calls for 20 minutes of re-teaching at the next session. Record the skill and one observation. Check it again with a fresh item at least a week later.

Complete answers for the week

Match the Day number, item ID, and student page before checking. Questions stay in the student workbook; these keys give every answer and the needed reasoning. Equivalent correct forms and sound alternative methods are welcome. An explanation item needs a reason, model, or valid check as well as its conclusion.

Area from a base and height

Day 043, student workbook pages 166-170

R1 [F-DIV] | student p. 166

Answer: $3\ \frac{1}{3}$

Reasoning: Rename the mixed number as an improper fraction, then count groups of three fourths.

Watch for $1\ \frac{7}{8}$: Multiplying by the divisor answers a different question.

R2 [D-MULT] | student p. 166

Answer: $7.35

Reasoning: 245 cents times three equals 735 cents.

R3 [E-EXP] | student p. 166

Answer: 9

Reasoning: Multiply 2 factors of 3.

Watch for 6: Multiplies the base by the exponent.

R4 [F-DIV] | student p. 166

Answer: 6

Reasoning: Rename each whole in thirds, then count groups of two thirds.

Watch for $2\ \frac{2}{3}$: Multiplying by the divisor answers a different question.

S [EXPLAIN] | student p. 166

Answer: Answers vary. A parallelogram rearranges into a rectangle. Two copies of a triangle make a parallelogram.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [G-AREA] | student p. 167

Answer: Area 15 square cm; perimeter 17 cm

Reasoning: Area counts square units: 6×2.5. Perimeter adds all four edge lengths.

Watch for 17 square cm: Uses perimeter as area.

P2 [G-PARA] | student p. 167

Answer: 28 square cm

Reasoning: Moving the end triangle makes a 7 by 4 rectangle.

Watch for 35 square cm: Uses the slanted side as height.

P3 [G-TRI] | student p. 167

Answer: 24 square cm both ways

Reasoning: Half of 6 × 8 equals half of 8 × 6.

Watch for 48 square cm: Finds the surrounding rectangle area.

P4 [G-TRI] | student p. 168

Answer: 60 square cm

Reasoning: The height splits the base into two 5 cm parts. Area = $\frac{1}{2}$ × 10 × 12.

Watch for 65 square cm: Uses a slanted side as height.

P5 [G-TRI] | student p. 168

Answer: 6 square cm

Reasoning: The base AB is 4 cm. The perpendicular height is 3 cm even though it lies outside.

Watch for 9 square cm: Uses the extended base length instead of AB.

P6 [G-TRI] | student p. 168

Answer: 6 square cm; no, the height must be perpendicular

Reasoning: Area = $\frac{1}{2}$ × 5 × 2.4. A slanted side usually gives the wrong distance between base and vertex.

Watch for 12 square cm: Omits the half.

F [G-TRI] | student p. 169

Answer: 2.5; 10 square cm

Reasoning: Use the worked example in the slides to check each step.

D [G-TRI] | student p. 169

Answer: The triangle is half the parallelogram: 18 square cm.

Reasoning: Counts both copies of the triangle. Place two matching paper triangles together.

E [G-TRI] | student p. 170

Answer: 48 square cm; height 8 cm

Reasoning: Half of the 12 by 8 surrounding rectangle is 48.

Watch for 60 square cm: Uses the side 10 as the height.

Nets and surface area

Day 044, student workbook pages 172-177

R1 [F-DIV] | student p. 172

Answer: 2 $\frac{1}{2}$; stories vary

Reasoning: Example: how many one-third-meter lengths fit in five sixths meter, allowing part of a length?

R2 [D-SUB] | student p. 172

Answer: 7.714

Reasoning: Write 10.000 and regroup through the zero columns.

R3 [E-ADD] | student p. 172

Answer: x = 14

Reasoning: Subtract 9 from both sides.

R4 [F-DIV] | student p. 172

Answer: 1 meters

Reasoning: Divide area by width. Check that length times width equals area.

Watch for $\frac{9}{16}$: Multiplying by the divisor answers a different question.

S [EXPLAIN] | student p. 172

Answer: Answers vary. A net unfolds every face. Surface area adds face areas in square units.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [G-COMP] | student p. 173

Answer: 42 square cm

Reasoning: Subtract the missing corner, or add two rectangles: 32 + 10 = 42. A third valid split gives 30 + 12.

Watch for 48 square cm: Counts the removed corner.

P2 [G-SA] | student p. 173

Answer: Surface area 52 square cm; volume 24 cubic cm

Reasoning: There are two faces of each size: 6, 8, and 12. Add twice their sum.

Watch for 24 square cm: Uses the volume formula.

P3 [G-SA] | student p. 173

Answer: Two triangles of area 24 each; rectangles of area 30, 40, 50; total 168 square cm

Reasoning: The two ends total 48. The three side rectangles total 120.

Watch for 144 square cm: Counts only one triangular end.

P4 [G-SA] | student p. 174

Answer: 96 square cm

Reasoning: The square base is 36. Four triangular faces each have area 15.

Watch for 156 square cm: Forgets the half on each triangular face.

P5 [G-SA] | student p. 174

Answer: 76 square cm

Reasoning: Add the missing bottom area, 8 square cm.

Watch for 68 square cm: Counts only the visible faces.

F [G-SA] | student p. 174

Answer: 3; 40 square cm

Reasoning: Use the worked example in the slides to check each step.

D [G-SA] | student p. 175

Answer: A closed cube has six faces: 6 × 9 = 54 square cm.

Reasoning: One hidden face was omitted. Label each face of the folded cube and net with matching letters.

M1 [F-DIV] | student p. 176

Answer: 12 pieces

Reasoning: Rename the total as thirty-six eighths. Divide into groups of three eighths.

M2 [N-DIV] | student p. 176

Answer: 13

Reasoning: Use long division and verify quotient × 24 equals the dividend.

M3 [E-RELATE] | student p. 176

Answer: (1, 2), (3, 6), (5, 10)

Reasoning: Multiply each input by 2.

E [G-SA] | student p. 177

Answer: 108 square cm

Reasoning: Two triangle ends total 12. Three rectangles total 24 + 32 + 40 = 96. Total 108.

Watch for 102 square cm: Leaves out one triangular end.

Study guide: Try it

Day 044, student workbook pages 178-181

1 [G-AREA] | student p. 181

Answer: 21 square cm

Reasoning: Multiply the two perpendicular side lengths.

2 [G-AREA] | student p. 181

Answer: 16 square cm

Reasoning: Multiply the two perpendicular side lengths.

3 [G-SA] | student p. 181

Answer: 62 square cm

Reasoning: Two faces of each type: 2 × (6 + 10 + 15).

Watch for 30: Uses volume.

4 [G-SA] | student p. 181

Answer: 66 square cm

Reasoning: Two faces of each type: 2 × (9 + 12 + 12).

Watch for 36: Uses volume.

5 [G-SA] | student p. 181

Answer: 88 square cm

Reasoning: Two faces of each type: 2 × (8 + 12 + 24).

Watch for 48: Uses volume.

Daily practice

Day 045, student workbook pages 185-186

1 [N-FACT] | student p. 185

Answer: 27

2 [N-FACT] | student p. 185

Answer: 63

3 [N-FACT] | student p. 185

Answer: 36

4 [N-FACT] | student p. 185

Answer: 81

5 [N-DIV] | student p. 185

Answer: 42

Reasoning: Thirty-two times forty is 1,280. Two more groups add 64.

6 [P-PERCENT] | student p. 185

Answer: 75 %

Reasoning: Compare the green count with the total. Rename the fraction as a rate per hundred.

7 [N-FACT] | student p. 185

Answer: 15

8 [G-PARA] | student p. 185

Answer: 48 square cm

Reasoning: Use base × perpendicular height: 6 × 8.

Watch for 60: Uses the slanted side.

9 [G-AREA] | student p. 186 | Built-in review: Area, surface area and volume

Answer: 9 square cm

Reasoning: Multiply the two perpendicular side lengths.

10 [G-PARA] | student p. 186 | Built-in review: Area, surface area and volume

Answer: 36 square cm

Reasoning: Use base × perpendicular height: 9 × 4.

Watch for 45: Uses the slanted side.

11 [G-TRI] | student p. 186 | Built-in review: Area, surface area and volume

Answer: 18 square cm

Reasoning: Half of 9 × 4 is 18.

Watch for 36: Omits the half.

12 [G-COMP] | student p. 186 | Built-in review: Area, surface area and volume

Answer: 39 square cm

Reasoning: Subtract the corner area from the full rectangle.

13 [G-TRI] | student p. 186

Answer: The triangle is half the parallelogram: 18 square cm.

Watch for 36 square cm: Counts both copies of the triangle.

Daily practice

Day 046, student workbook pages 187-188

1 [F-EQUIV] | student p. 187

Answer: 0.5; 50%

2 [F-EQUIV] | student p. 187

Answer: 0.25; 25%

3 [F-EQUIV] | student p. 187

Answer: 0.75; 75%

4 [F-EQUIV] | student p. 187

Answer: 0.125; 12.5%

5 [E-EXP] | student p. 187

Answer: 9

Reasoning: Multiply 2 factors of 3.

Watch for 6: Multiplies the base by the exponent.

6 [F-DIV] | student p. 187

Answer: $2\frac{1}{2}$; stories vary

Reasoning: Example: how many one-third-meter lengths fit in five sixths meter, allowing part of a length?

7 [N-FACT] | student p. 187

Answer: 12

8 [G-SA] | student p. 187

Answer: 66 square cm

Reasoning: Two faces of each type: 2 × (9 + 12 + 12).

Watch for 36: Uses volume.

9 [G-SA] | student p. 188 | Built-in review: Area, surface area and volume

Answer: 62 square cm

Reasoning: Two faces of each type: 2 × (6 + 10 + 15).

Watch for 30: Uses volume.

10 [G-SA] | student p. 188 | Built-in review: Area, surface area and volume

Answer: 60 square cm

Reasoning: Two ends give 12 square cm. The three rectangles give 48.

Watch for 54: Omits one triangular end.

11 [G-SA] | student p. 188 | Built-in review: Area, surface area and volume

Answer: 40 square cm

Reasoning: Add the square base and four half-base-times-height triangles.

12 [G-SA] | student p. 188 | Built-in review: Area, surface area and volume

Answer: 24 square cm

Reasoning: Six faces are needed. Each has area 4.

Watch for 20: Leaves out the hidden face.

13 [G-SA] | student p. 188

Answer: A closed cube has six faces: 6 × 9 = 54 square cm.

Watch for 45 square cm: One hidden face was omitted.

Daily practice

Day 047, student workbook pages 189-190

1 [N-SCALE] | student p. 189

Answer: 92

2 [N-SCALE] | student p. 189

Answer: 9

3 [N-SCALE] | student p. 189

Answer: 19

4 [N-SCALE] | student p. 189

Answer: 27

5 [G-COMP] | student p. 189

Answer: 38 square cm

Reasoning: Subtract the corner area from the full rectangle.

6 [G-PARA] | student p. 189

Answer: 24 square cm

Reasoning: Use base × perpendicular height: 8 × 3.

Watch for 40: Uses the slanted side.

7 [E-ADD] | student p. 189

Answer: x = 14

Reasoning: Subtract 9 from both sides.

8 [D-MULT] | student p. 189

Answer: 0.56; yes

Reasoning: Eight tenths of seven tenths covers fifty-six hundredths.

9 [F-MULT] | student p. 190

Answer: $\frac{3}{5}$

Reasoning: An area model shows 12 of 20 equal parts, or three fifths.

10 [F-COMP] | student p. 190

Answer: $\frac{2}{5}$, 0.45, $\frac{1}{2}$

Reasoning: Rename two fifths as 0.40 and one half as 0.50.

11 [N-FACT] | student p. 190

Answer: 10

12 [G-VOL] | student p. 190

Answer: 30 cm^3

13 [G-AREA] | student p. 190 | Built-in review: Weekly puzzle

Answer: 5 cm and 7 cm

Reasoning: The sides sum to 12. Test factor pairs of 35: 1 and 35 fail; 5 and 7 sum to 12.

Watch for 3 cm and 8 cm: Checks neither condition completely.

Daily practice

Day 048, student workbook pages 191-192

1 [N-EST] | student p. 191

Answer: 900

2 [N-EST] | student p. 191

Answer: 10

3 [N-EST] | student p. 191

Answer: About 100

4 [N-EST] | student p. 191

Answer: About 20

5 [G-COMP] | student p. 191

Answer: 38 square cm

Reasoning: Subtract the corner area from the full rectangle.

6 [G-PARA] | student p. 191

Answer: 24 square cm

Reasoning: Use base × perpendicular height: 8 × 3.

Watch for 40: Uses the slanted side.

7 [G-SA] | student p. 191

Answer: 62 square cm

Reasoning: Two faces of each type: 2 × (6 + 10 + 15).

Watch for 30: Uses volume.

8 [E-EXPR] | student p. 191

Answer: 8b; 32 beads

Reasoning: 8 equal groups of b give 8b.

9 [E-DIST] | student p. 192 | Built-in review: Expressions and equations

Answer: 3x + 18

Reasoning: Multiply x and 6 by 3.

Watch for 3x + 6: Multiplies only the first term.

10 [E-EXP] | student p. 192 | Built-in review: Expressions and equations

Answer: 16

Reasoning: Multiply 4 factors of 2.

Watch for 8: Multiplies the base by the exponent.

11 [E-MULT] | student p. 192 | Built-in review: Expressions and equations

Answer: x = 9

Reasoning: Divide both sides by 6.

Watch for 48: Subtracts instead of dividing.

12 [E-RELATE] | student p. 192 | Built-in review: Expressions and equations

Answer: (0, 0), (2, 8), (3, 12)

Reasoning: Multiply each input by 4.

13 [G-SA] | student p. 192

Answer: 120 square cm

Reasoning: Two ends give 12 square cm. The three rectangles give 108.

Watch for 114: Omits one triangular end.

14 [G-SA] | student p. 192

Answer: A closed cube has six faces: 6 × 9 = 54 square cm.

Watch for 45 square cm: One hidden face was omitted.

Weekly check

Day 049, student workbook pages 193-194

1 [G-COMP] | student p. 193

Answer: 38 square cm

Reasoning: Subtract the corner area from the full rectangle.

2 [G-PARA] | student p. 193

Answer: 24 square cm

Reasoning: Use base × perpendicular height: 8 × 3.

Watch for 40: Uses the slanted side.

3 [G-SA] | student p. 193

Answer: 62 square cm

Reasoning: Two faces of each type: 2 × (6 + 10 + 15).

Watch for 30: Uses volume.

4 [G-SA] | student p. 193

Answer: 96 square cm

Reasoning: Add the square base and four half-base-times-height triangles.

5 [E-EVAL] | student p. 193

Answer: 11

Reasoning: 4 × 2 + 3 = 11.

6 [D-DIV] | student p. 194

Answer: 0.375

Reasoning: Continue the long division into tenths, hundredths, and thousandths until the remainder is zero.

7 [E-MULT] | student p. 194

Answer: x = 3.2

Reasoning: Divide both sides by 1.5; then check the product.

8 [F-MULT] | student p. 194

Answer: $\frac{3}{5}$

Reasoning: An area model shows 12 of 20 equal parts, or three fifths.

9 [F-COMP] | student p. 194

Answer: $\frac{2}{5}$, 0.45, $\frac{1}{2}$

Reasoning: Rename two fifths as 0.40 and one half as 0.50.

10 [G-TRI] | student p. 194

Answer: The triangle is half the parallelogram: 18 square cm.

Watch for 36 square cm: Counts both copies of the triangle.